

Technology Stack

Date	3 July 2026
Team ID	XXXXXX
Project Name	Rising Waters - Flood Prediction System
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	<i>HTML5, CSS3, JavaScript</i>	<i>Used to build a responsive, interactive, and user-friendly web interface for entering weather parameters and displaying flood prediction results.</i>

architecture and provide a brief justification for your choice.



3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	<i>CSV Dataset & Pickle (.pkl) Model File</i>	<i>The historical weather dataset is stored in CSV format for training, and the trained Decision Tree model is stored as a .pkl file for fast prediction without retraining.</i>
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	<i>IBM Cloud (Planned)</i>	<i>IBM Cloud provides scalable and reliable deployment, making the application accessible from any location.</i>
5	Version Control & CI/CD	<i>Git & GitHub</i>	<i>Git tracks code changes, while GitHub enables collaboration, version control, and project management among team members.</i>

	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	<i>Scikit-learn, Pandas, NumPy, Matplotlib, Joblib</i>	<i>Scikit-learn is used to build the Decision Tree model, Pandas and NumPy handle data processing, Matplotlib is used for data visualization, and Joblib is used to save and load the trained model efficiently.</i>

Technology Stack Details

